

OOP

Easy

- Procedural Programming VS OOP?
- What is a class and what is an object?
- What is Main Pillars of OOP?
- Explain inheritance with a real-world example
- What is the difference between data hiding and encapsulation?

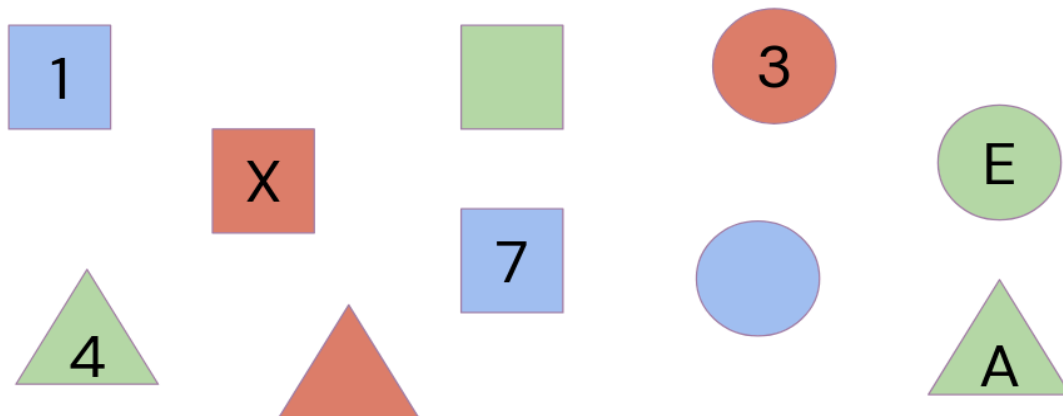
Medium

- Types of Inheritance in OOP
- Abstract Class VS Interface
- What is the Polymorphism?
- Overloading VS Overriding
- Overloading VS Generic Programming (like templates in C++)

Hard

- Types of Relationships in OOP? Difference between them?
- Can a class inherit from multiple classes in Java/C#? How can you achieve a similar outcome?
- What is the "Diamond Problem"? How do languages like C++, Java handle it
- What is the difference between Coupling and Cohesion?

Arrange the content in below image



Note: should be aware of separation of concerns

Problem Solving

Easy

- Given a string, write a function to check if it is a palindrome
- Given an array of integers and a target number, return the indices of the two numbers that add up to the target.
 - $O(n^2)$ -> easy
 - $O(n \log(n))$ -> medium
 - $\sim O(n)$ -> hard (using hashing like unordered set)

Medium

- Can we sort an array of n elements in $O(n)$? explain how, and limitation of it
- Given a string, find the first non-repeating character in it and return its index. If it doesn't exist, return -1.
 - $O(n^2)$ -> easy
 - $O(n)$ using freq array -> medium
 - $O(n)$ using visited array contain indices -> hard

Power of 2?

Q: How to know that a number is power of 2 in Time $O(1)$.

Hint 1: Figure it out using the binary representation of the number.

Hint 2: What happens to the binary representation of any number after subtracting 1?

El King and Amira

Topics: <math,number theory>

Q: Given integer $X \rightarrow$ print YES if X has odd number of divisors, otherwise NO

Constraints: 1 second , 256 mb, $(1 \leq X \leq 1e15)$

Sample 1: Input $\rightarrow$ 12 output $\rightarrow$ NO

Sample 2: Input $\rightarrow$ 36 output $\rightarrow$ YES

First Solution $\rightarrow$ Based on simple observation (if the X Perfect Square Number then it has odd number of divisors) (e.g. $\text{div}(4) = \{1,2,4\}$ $O(\log(n))$)

Second Solution $\rightarrow$ Iterate from 2 to $\sqrt{X}$ and count the divisors i and n/i

- Sure that the counter i is long long $\rightarrow$ To avoid intermediate overflow
- Time Complexity : $O(\sqrt{X})$ or $(O(\sqrt{X}) * \log(x))$ when use the sqrt function)

Jump Game (BFS)

Given an array of non-negative integers `arr`, you are initially positioned at start index of the array. When you are at index `i`, you can jump to `i + arr[i]` or `i - arr[i]`, check if you can reach any index with value 0. In $O(n)$ time and memory.

Notice that you can not jump outside of the array at any time.

Sample: `arr = [4,2,3,0,3,1,2]`, `start = 5`

Output : true

All Possibles:

index 5 -> index 4 -> index 1 -> index 3

index 5 -> index 6 -> index 4 -> index 1 -> index 3

Constraints:

$1 \leq \text{arr.length} \leq 5 \times 10^4$

$0 \leq \text{arr}[i] < \text{arr.length}$

$0 \leq \text{start} < \text{arr.length}$

Data Structures

Easy

- Discuss Usage of DS
- Linear VS Non-linear DS
- Stack VS Queue (by Logic)
- Stack that implemented with ONLY SINGLE Queue , what are the push and pop functions?Why?
- Complexities of Insertion,Deletion, Access (front,back,middle)

Medium

- What is the non-linear ds that based on linear DS?
- Discuss the Deletion of BST
- When the operation of BST will be $O(n)$? (Draw the Example)
- All DS based on Physical DS
- Physical VS Logical DS
- What is the Serialization concept?
- What is the All possible binary trees with 3 nodes? What is the formula?

Hard

- What is the Type of AVL Tree?
- What is the balance factor in AVL?
- Discuss the Rotation in AVL.
- Trie Basic Problem
- Unique Anagrams (Basic on Hashing)

Design Principles

- What is the helper principles that used to make effective OOP?
- What is the difference between Design Principles and Design Patterns?
- What is the Single Responsibility Principle? (critical question)
- What is the difference between Dependency Inversion and Dependency Injection?
- Examples on SoC in SWE

Design Patterns

- Creation vs Structural vs Behavioural and examples on them
- Can you describe the Observer pattern? What is a common, real-world example
- When would you declare a constructor as private? What design pattern does this facilitate?

Database

- Database VS Files
- What is the difference between primary key and a unique constraint?
- Explain the difference between Cross Join, Inner Join, Left Join, Right Join
- What is database normalization and why is it important?
- What is the difference between the SQL statements delete, truncate, drop?
- Difference between Subqueries and CTEs